

Financial Accounting

Executive MBA

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Lecture 8: Intercompany Equity Investments

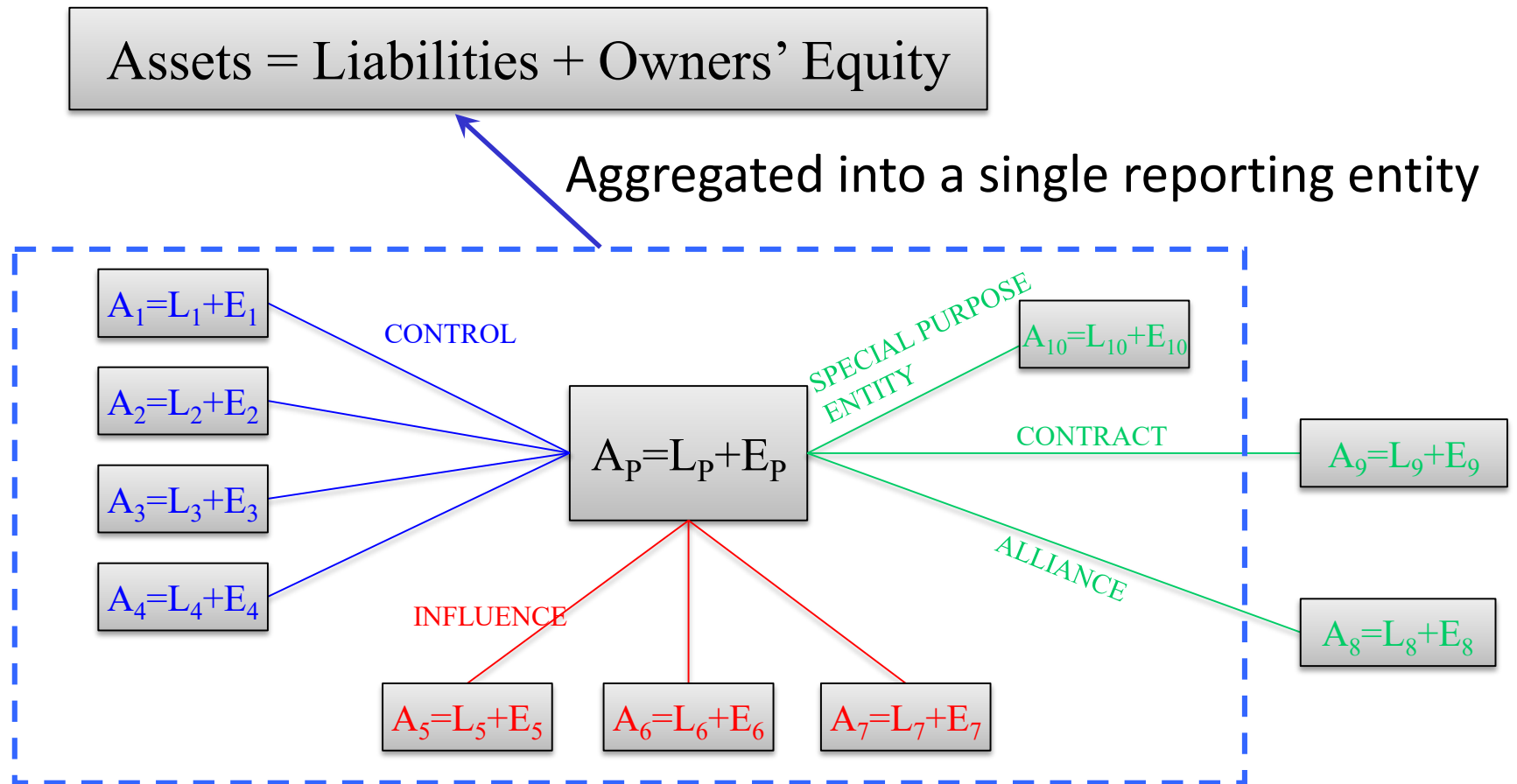


The University of Chicago Booth School of Business

How should the following transactions be recognized?

1. Berkshire Hathaway currently owns about 5.6% of Apple's voting common stock. The goal of Berkshire's investment is to earn a high rate of return from Apple's investment.
2. The Coca-Cola Company (Coke) currently owns about 23% of the voting rights of Coca-Cola Hellenic Bottling Company. Coke HBC is a key bottler for Coke and Coke holds several board seats of Coca-Cola HBC.
3. LVMH acquired a 51% equity stake in Bulgari SpA in 2011 with the 49% equity stake maintained by the Bulgari family. LVMH's stake allowed it to make the major operating and financial decisions for Bulgari SpA.
4. Two entities X and Y set up a joint venture to work on a new drug. Both entities will share all profits from the joint venture equally but Y will absorb 80% of any losses from the joint venture.

Determining the boundary around operations under a firm's influence and/or control:



Companies account for intercompany investments based on

1. the type of security (debt or equity) and
2. their intent with respect to the investment.

Type of Security	Management Intent	Valuation Approach
Debt	No plans to sell	Amortized cost
	Plan to sell	Fair value
Equity	Plan to sell	Fair value
	Exercise influence without control	Equity method
	Legal Control	Consolidation

Financial Reporting for Debt Investments

Category	Unrealized Holding		
	Valuation	Gains or Losses	Other Income Effects
Held-to-maturity	Amortized cost	Not recognized	Interest when earned; gains and losses from sale.
Trading	Fair value	Recognized in net income	Interest when earned; gains and losses from sale.
Available-for-sale	Fair value	Recognized as other comprehensive income and as separate component of stockholders' equity	Interest when earned; gains and losses from sale.

Debt classified as Held-to-Maturity (HTM)

- Classify a debt security as **held-to-maturity** only if management has **both**
 1. the positive intent and
 2. the ability to hold securities to maturity.
- Measured at amortized cost, not fair value.
- Amortize any premium or discount using the **effective-interest method**.

HTM: An Illustration

Firm invests in a two-year bond with face value of \$1,000 with a coupon rate of 6% and priced to yield 10% issued on 31/12/2024. At 31/12/2024, firm believes that it has both the intent and ability to hold debt investment until maturity, i.e., until 31/12/2026. Coupon interest is received annually on every Dec 31st.

- Coupon interest = \$ 1,000 x 6% = \$60 every year
- Issue price of bond = $\$60 \times PVANN(2,10\%) + \$1,000 \times PV(2,10\%)$
= $\$60 \times 1.74 + 1,000 \times 0.8265$
= $104.0 + 826.5 = \$930.5$

Recognizing investment in bond on 31/12/2024

Investment in bonds	\$930.5	
	Cash	\$930.5

HTM : An Illustration

Recognizing interest income during fiscal 2025 and 2026

Investment in Bonds	\$33.1	
Cash	\$60.0 (= 6%*1,000)	
	Interest Income	93.1
Investment in Bonds	\$36.4	
Cash	\$60.0 (= 6%*1,000)	
	Interest Income	96.4

Receipt of principal at maturity, i.e., at 31/12/2026

Cash	\$1,000	
Investment in Bonds		\$1,000

Debt classified as Trading

- Companies report **trading securities** if management has intention to sell before maturity:
 - Measured at fair value every period, with
 - unrealized holding gains and losses reported as part of *net income* where holding gain or loss is the net change in fair value of a security from one period to another.
- In practice, debt securities that management intends to *sell in the near term* are classified as trading.

Debt classified as Available for sale (AFS)

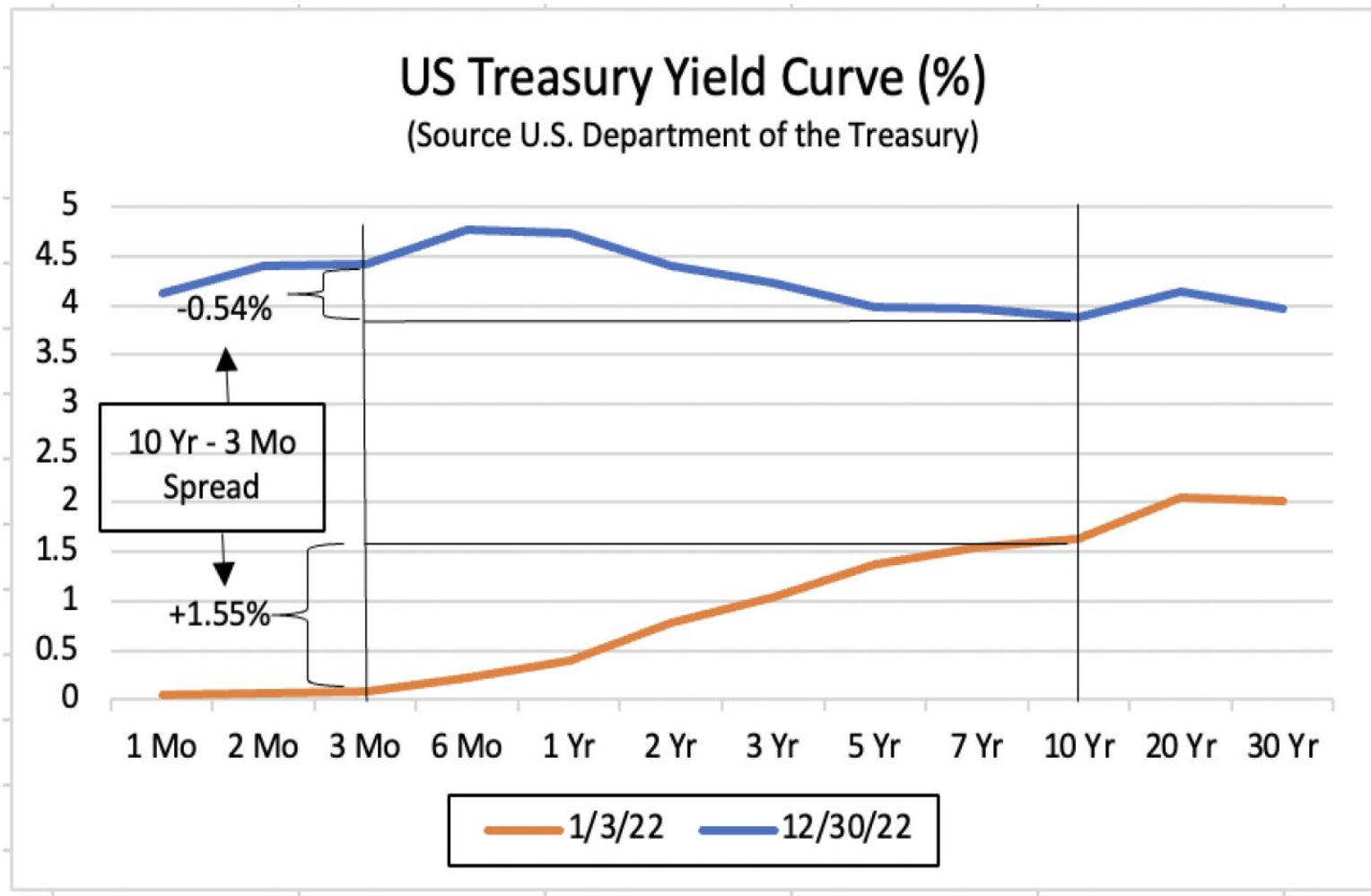
- Companies report **AFS** if management has intention to sell before maturity:
 - Measured at fair value every period, with
 - unrealized holding gains and losses unrealized holding gains and losses reported as *other comprehensive income* (OCI), a separate component of stockholders' equity until realized.
 - In practice, debt securities are *not classified as either HTM or Trading* are classified as AFS.

Silicon Valley Bank (SVB): AFS vs. HTM

SVB FINANCIAL GROUP AND SUBSIDIARIES CONSOLIDATED BALANCE SHEETS

(Dollars in millions, except par value and share data)	December 31,	
	2022	2021
Assets		
Cash and cash equivalents	\$ 13,803	\$ 14,586
Available-for-sale securities, at fair value (cost of \$28,602 and \$27,370, respectively, including \$530 and \$61 pledged as collateral, respectively)	26,069	27,221
Held-to-maturity securities, at amortized cost and net of allowance for credit losses of \$6 and \$7 (fair value of \$76,169 and \$97,227, respectively)	91,321	98,195
Non-marketable and other equity securities	2,664	2,543
Total investment securities	120,054	127,959
Loans, amortized cost	74,250	66,276
Allowance for credit losses: loans	(636)	(422)
Net loans	73,614	65,854
Premises and equipment, net of accumulated depreciation and amortization	394	270
Goodwill	375	375
Other intangible assets, net	136	160
Lease right-of-use assets	335	313
Accrued interest receivable and other assets	3,082	1,791
Total assets	\$ 211,793	\$ 211,308
Liabilities and total equity		
Liabilities:		
Noninterest-bearing demand deposits	\$ 80,753	\$ 125,851
Interest-bearing deposits	92,356	63,352
Total deposits	173,109	189,203
Short-term borrowings	13,565	71
Lease liabilities	413	388
Other liabilities	3,041	2,467
Long-term debt	5,370	2,570
Total liabilities	195,498	194,699
Commitments and contingencies (Note 21 and Note 26)		
SVBFG stockholders' equity:		
Preferred stock, \$0.001 par value, 20,000,000 shares authorized; 383,500 and 383,500 shares issued and outstanding, respectively	3,646	3,646
Common stock, \$0.001 par value, 150,000,000 shares authorized; 59,171,883 and 58,748,469 shares issued and outstanding, respectively	—	—
Additional paid-in capital	5,318	5,157
Retained earnings	8,951	7,442
Accumulated other comprehensive income (loss)	(1,911)	(9)
Total SVBFG stockholders' equity	16,004	16,236
Noncontrolling interests	291	373
Total equity	16,295	16,609
Total liabilities and total equity	\$ 211,793	\$ 211,308

Interest rate environment in 2022



Impact of rising interest rates on banks' business model

In early 2022, banks were sitting on investments that they originated in the sub-2% yield environment and funding them primarily with demand deposits on which they pay little to no interest. During 2022, the Fed raised interest rates. What could go wrong?

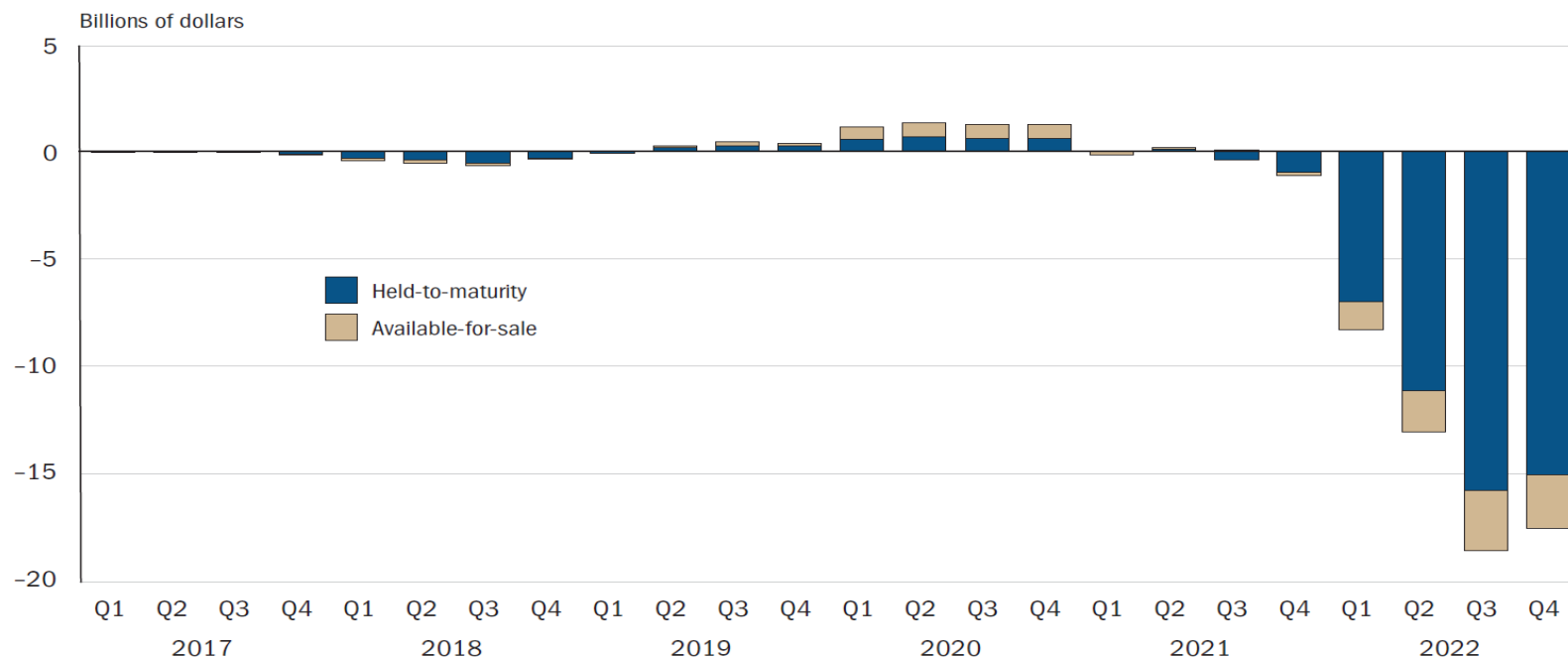
- a) Depositors withdraw their funds to earn higher interest rates elsewhere.
- b) Banks are forced to pay higher deposit account interest rates to retain depositors.
- c) Uninsured depositors withdraw funds after realizing that the value of banks' investments has dropped and may not be enough to pay everyone back. This forces banks to sell their investments and realize any associated losses and/or seek higher cost financing.
- d) Banks' depositor bases are concentrated in areas where the customers' businesses are struggling causing a large amount of deposits to be withdrawn. This forces banks to sell their investments and realize any associated losses and/or seek higher cost financing.
- e) All of the above.

Silicon Valley Bank (SVB)

- SVB's Leverage at 31/12/2022 = Liabilities/Equity = $195,498/16,295 = 12.0$.
- At 31/12/2022, HTM securities had a carrying value of \$91,321 millions and fair value = \$76,169 millions.
 - Equity of SVB based on *carrying value* of HTM portfolio = \$16,295 millions but equity based on *fair value* of HTM portfolio = \$1,143 millions.
- % of funding from deposits at 31/12/2022 = $173,109/195,498 = 89\%$
- % of funding from noninterest-bearing deposits at 31/12/2022 = $80,753/173,109 = 47\%$ while % of funding from noninterest bearing deposits at 31/12/2021 = $125,851/189,203 = 67\%$ so that the reliance on interest-bearing deposits went up during fiscal 2022.

SVB's Unrealized Gains/Losses on Securities Portfolio

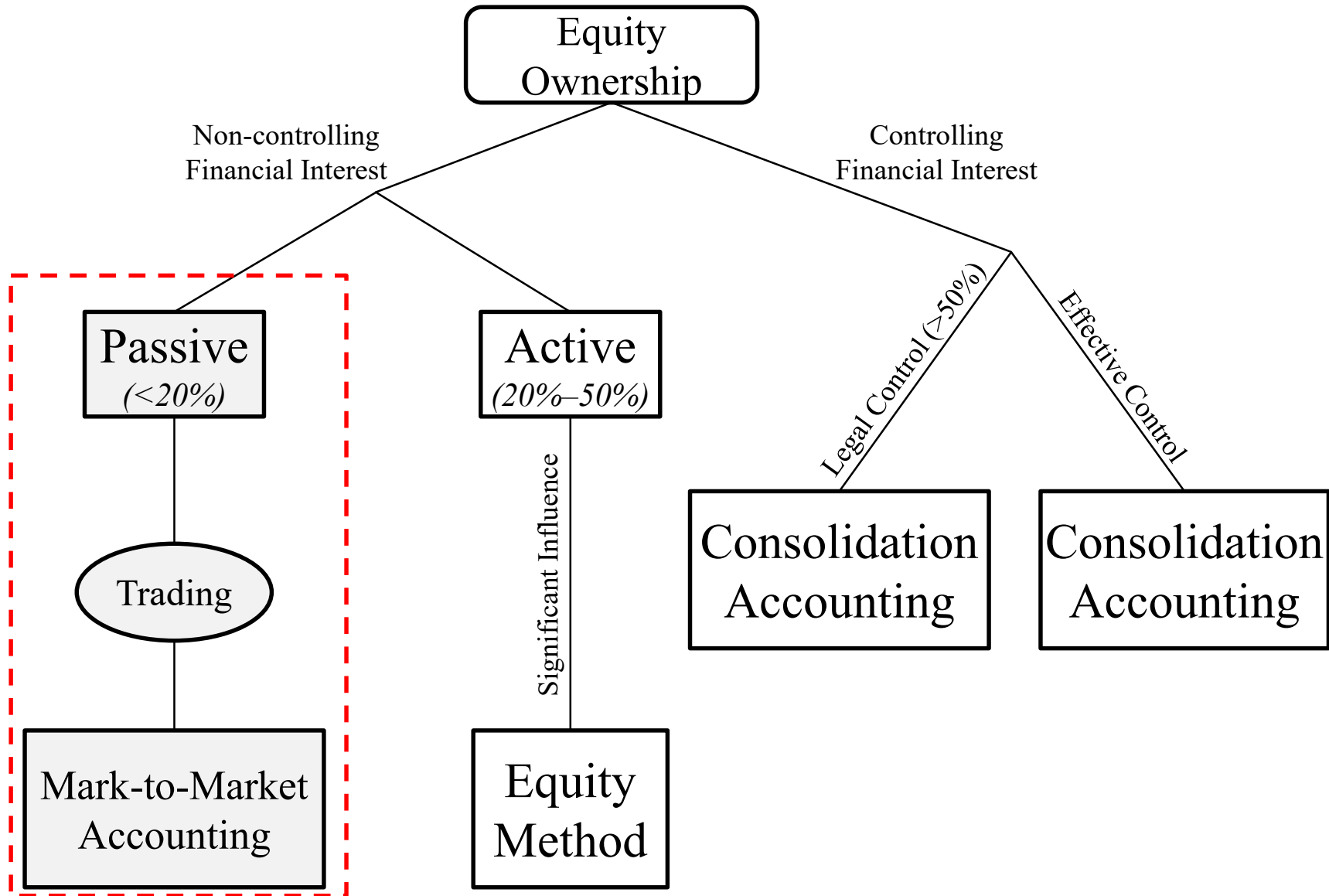
Figure 9. Estimated unrealized gains (losses) on SVBFG's investment portfolio securities



Note: Estimated unrealized losses on securities calculated as: (held-to-maturity fair value less held-to-maturity amortized cost) + (available-for-sale fair value less available-for-sale amortized cost). Estimates do not reflect losses related to available-for-sale securities that were transferred to held-to-maturity and do not reflect hedging impacts or tax consequences.

Source: FR Y-9C.

Financial Reporting for Equity Investments



Equity Investments: Passive

- Acquiring firm has **no significant influence** over acquired shareholder's operating and financial policies.
 - In the absence of any other information, the presumption is that ownership of less than 20% of voting rights usually implies lack of significant influence and investment is viewed as passive.
- Equity investments are classified as *trading* securities:
 - If trading securities have a readily available market price, securities are marked to market at each financial reporting date and any unrealized holding gains and losses reported as part of net income.
 - Dividends received from the corporate investee are recorded as dividend income for GAAP purposes.
 - Realized losses or gains are recorded when the securities are sold.
 - Earnings of investee are not recognized by the investor.

Trading Securities

- Any *unrealized gains or losses* from marking trading securities to market flow through the ***income statement*** to capture the speculative gains or losses from marking the securities up or down to market:
 - For tax purposes, these securities are not marked to market. These unrealized gains and losses recognized on the income statement are viewed as temporary tax differences. The difference between book value vs. tax basis of trading investments results in the recognition of deferred income taxes.
- When trading securities are sold, realized gains or losses flow through the income statement and previously recognized deferred taxes reverse.
- Impact of trading securities on income statement:
 - Dividend income from Investee,
 - Unrealized gains/losses from marking to market, and
 - Realized gains/losses from sales.

Passive Investments: An Illustration

- *Booth* Corp (*Booth*) was incorporated in Chicago, IL at the end of fiscal year (FYE 0) with an initial balance sheet consisting of the following:

Booth's Initial Balance Sheet: FYE0	
Cash	\$200,000
Equity—C/S	\$200,000

During FY1, *Booth* purchases a portfolio of equity securities. The portfolio is sold during FY2. The transactions associated with the portfolio are given below. All purchases and sales are in cash.

Security	Purchases during FY1	Market price at FYE 1	Sales during FY2
<i>A</i>	\$48,000	\$25,000	\$15,000
<i>B</i>	29,000	55,000	70,000
<i>Total</i>	\$77,000	\$80,000	\$85,000

Financial Reporting: Passive Investor

Additional Information:

- Both investments were passive. Being equity securities, *Booth* classified both securities *A* and *B* as trading securities (TS).
- For tax purposes, the securities are not marked to market. Instead, taxable gains and tax-deductible losses are recognized when securities are sold for cash. The difference between book value and tax basis of securities from marking-to-market accounting would be treated as temporary book-tax differences.

Requirements: Prepare *Booth*'s major financial statements for fiscal years 1 and 2. *Booth* treats the securities portfolio as an *investing* (instead of an operating) activity. *Booth*'s statutory tax rate is 21% and any taxes owed during a fiscal year are paid in cash during the same fiscal year.

Financial Reporting Implications

- Acquisition of portfolio during FY1:

TS-A	\$48,000	
	Cash	\$48,000
TS-B	\$29,000	
	Cash	\$29,000

- Marking to market the portfolio during FY1:

Unrealized Loss (I/S)	\$23,000	
	TS-A	\$23,000
TS-B	\$26,000	
	Unrealized Gain (I/S)	\$26,000

MTM and its Tax Implications

- Entries to reflect the tax impact of unrealized gains/losses for FY1:

DTA (TS-A)	\$4,830	
	Income tax benefit	\$4,830
Income tax expense	\$5,460	
	DTL (TS-B)	\$5,460

- Combining the two entries above, we get:

Income tax expense-net	\$630	
	DTL-net	\$630

- Closing Entry at end of FY1:

Unrealized gain-net	\$3,000	
	Income tax expense-net	\$630
	Retained Earnings	2,370

Booth's Financial Statements: FY1

Income Statement

Operating Income	0
Other Income-net	\$ 3,000
Income before taxes	\$ 3,000
–Income tax expense	–630
Net Income	\$ 2,370

Tax Return

Taxable Income	\$0
x Statutory Tax Rate	21%
Taxes paid	\$0

Cash Flow Statement

Net Income	\$ 2,370
Net unrealized gains	–3,000
Change in Deferred Taxes-net	630
CFO	\$ 0
CFI	–\$77,000
CFF	\$ 0
Net Δ in Cash	–77,000
+ Beginning Cash	\$ 200,000
Ending Cash	\$123,000

Balance Sheet

Cash	\$ 123,000
Securities	80,000
Assets	\$ 203,000
DTL-net	630
C/S	\$ 200,000
Retained Earnings	2,370
Equity	\$202,370
Liabilities & Equity	\$203,000

Financial Reporting Implications of Sales

- Sale of TS-*A* in FY2:

Cash	\$15,000	
Realized Loss	10,000	
	TS- <i>A</i>	\$25,000

- Sale of TS-*B* in FY2:

Cash	\$70,000	
	TS- <i>B</i>	\$55,000
	Realized Gain	15,000

Tax Implications of Sales

- The tax deductible loss on sale of TS-A is $\$48,000 - 15,000 = \$33,000$ while the taxable gain on sale of TS-B is $\$70,000 - 29,000 = \$41,000$.
- The *taxable income* from selling the portfolio in FY2 is therefore $\$41,000 - 33,000 = \$8,000$ resulting in taxes owed during FY2 of $\$8,000 \times 0.21 = \$1,680$.
- The *book income* from selling the portfolio in FY2 is the realized book loss of \$10,000 on TS-A netted against the realized book gain of \$15,000 on TS-B resulting in a book income of \$5,000 and an income tax expense of $\$5,000 \times 0.21 = \$1,050$.
- The net DTL of \$630 set up in FY1 fully reverses in FY2 to reconcile the difference between income tax expense of \$1,050 based on the book income of \$5,000 vs. taxes owed of \$1,680 based on taxable income of \$8,000:

Income Tax Expense	$\\$5,000 \times 0.21 = \\$1,050$	
DTL-net	630	
	Cash taxes (calculated) [Taxable income of \$8,000 x 0.21]	\$1,680

Financial Reporting: Passive Investor

- Closing Entry at end of fiscal year 2:

Realized Gain-net (TS-A & TS-B)	\$5,000	
	Income Tax Expense	\$1,050
	Retained Earnings	3,950

Booth's Financial Statements: FY2

Income Statement

Operating Income	0
Other income-net	\$ 5,000
Income before taxes	\$ 5,000
– Income tax Expense	–1,050
Net Income	\$3,950

Tax Return

Taxable Income	\$8,000
x Statutory Tax Rate	21%
Taxes paid	\$1,680

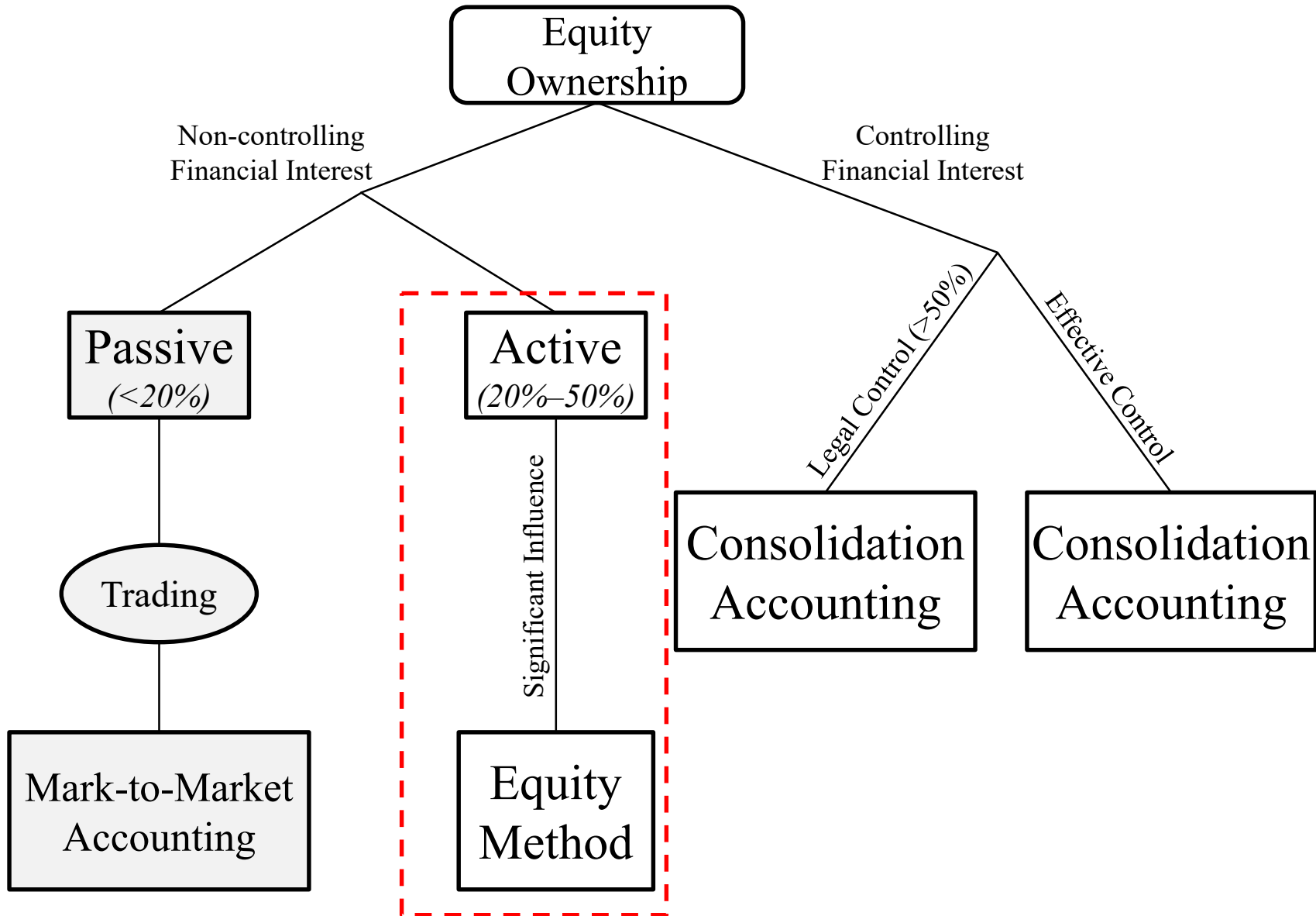
Cash Flow Statement

Net Income	\$ 3,950
Realized Gain-net	–5,000
Change in DTL-net	–630
CFO	– \$1,680
CFI	85,000
CFF	0
Net Δ in Cash	\$83,320
+ Beginning Cash	123,000
= Ending Cash	\$206,320

Balance Sheet

Cash	\$206,320
Securities	0
DTA	0
Assets	\$206,320
DTL	0
C/S	\$200,000
Retained Earnings	6,320
Equity	\$206,320
Liabilities + Equity	\$206,320

Financial Reporting for Active Equity Investments



Equity Investment: Active

- Active investment: acquirer company's investment is viewed as having *significant influence* over acquired company's operating and financial policies. These inter–corporate investments are usually made for strategic reasons including:
 - **Prelude to acquisition:** Significant ownership can allow investor company to gain seats on the board from which it can learn about the investee company.
 - **Strategic Alliance:** One example of a strategic alliance is an investment company that provides inputs for the investor's production process. This relationship is closer than the usual supplier-buyer relationship, often because the investor company provides trade secrets or technical know-how of its production process.
 - **Pursuit of research and development:** Many research activities in the pharmaceutical, software and oil and gas companies are conducted jointly. These joint ventures allow various firms to share risks or capital investment.

When is an Equity Investment Active?

- If investment allows an investor to exert *significant influence* over *financing or operating policies* of investee, then investment is viewed as active and should be accounted for using the *equity method*:
 - Some indicators of significant influence:
 - Representation on the Board.
 - Material intercompany transactions.
 - Greater ownership than other shareholders.
 - In practice, ownership levels of at least 20% and below 50% (direct and indirect) of voting rights over the investee presume significant influence.

Who Uses the Equity Method?

- Corporate investors who own 20%–50% of voting rights over investee.
- Corporate investors who share control in a joint venture.
- Corporate investors who own less than 20% of the voting common stock, but can still exercise a significant influence.
- Corporate investors who own over 50% of the voting common stock in a subsidiary, but have temporary or no control over the subsidiary:
 - Subsidiary is under bankruptcy reorganization.
 - Subsidiary has minority shareholders with veto rights.

Equity Method: Financial Reporting

- Under the equity method, the investor records the initial investment at cost:
 - The carrying value of the investment is then increased for the investor's pro-rata share of the investee's earnings or decreased for the investor's pro-rata share of the investee's loss.
 - Any dividends from investee are viewed as a *liquidation or recovery* of the investment in the investee and reduces the carrying value of the investment.
- At acquisition, the cost of investment typically exceeds book Value of Net Assets (BVNA) or equity acquired. Why?
 - Excess of cost over book value of net assets acquired = Net Asset Step-ups + Implicit Goodwill

Equity Method: A Simple Illustration

- Acquirer Company acquired 50% of Target Company's *outstanding* voting common stock by issuing Acquirer stock with a market price \$15,000 millions on 1/1/2024. The acquisition allows the Acquirer to exercise *significant influence but not control* over the Target's major policies. The balance sheets of Acquirer and Target on 1/1/2024, right before acquisition, are shown on the next page.
- An analysis of Target's balance sheet revealed that all recognized assets and liabilities had carrying (book) values equal to their fair (market) values at the acquisition date.

Equity Method: A Simple Illustration

- The standalone balance sheets of Acquirer and Target Inc., *right before* the stock acquisition were as follows:

Balance Sheets at 1/1/2024 (\$ amounts in thousands)	Acquirer Inc. (A)	Target Inc. (T)
Cash and receivables	\$40,000	\$10,000
Inventories	20,000	25,000
PP&E	60,000	10,000
Total Assets	\$120,000	\$45,000
Liabilities	60,000	20,000
Equity	60,000	25,000
Total Liabilities and Equity	\$120,000	\$ 45,000

Equity Method: A Simple Illustration

To recognize acquisition of 50% of *T*'s Equity on 1/1/2024

Investment in <i>T</i>	\$15,000,000	
	Equity- <i>A</i>	\$15,000,000

Balance Sheets post acquisition (\$ amounts in thousands)	<i>A</i>	<i>T</i>
Cash and receivables	\$40,000	\$10,000
Inventories	20,000	25,000
PP&E	60,000	10,000
<i>Investment in Target</i>	<i>15,000</i>	--
Total Assets	\$135,000	\$45,000
Liabilities	60,000	20,000
<i>Equity</i>	<i>75,000</i>	25,000
Total Liabilities and Equity	\$135,000	\$ 45,000

Equity Method: A Simple Illustration

Why did Acquirer pay \$15,000,000 for the target's equity that has a carrying value of $50\% \times \$25,000,000 = \$12,500,000$?

Fair Value of 50% of <i>T</i>'s Equity = Cost of Equity Acquired (\$ amounts in thousands)	\$ 15,000
– Carrying Value of Net Assets Acquired = $50\% \times \$25,000$	12,500
= Excess cost over carrying value of net assets acquired	2,500
– Adjust <i>T</i> 's net assets to their respective fair values	0
= Excess of cost over <i>A</i> 's share of net assets of <i>T</i> or equivalently, <i>Implied Goodwill</i>	\$ 2,500

Equity Method: A Simple Illustration

- During fiscal 2024, *T* recognizes net income of \$10,000,000:

To recognize *A*'s share of *T*'s income ($50\% \times \$10,000,000$):

Investment in <i>T</i>	\$5,000,000	
	Equity Income in <i>T</i>	\$5,000,000

- T* pays cash dividends of \$1,000,000 to all its shareholders:

To recognize *A*'s share of *T*'s cash dividends of $50\% \times \$1,000,000$:

Cash	\$500,000	
	Investment in <i>T</i>	\$500,000

Movements in Investment Account: Fiscal 2024

Investment in T (Asset)	
\$15,000,000 (acquisition cost)	
	\$500,000 (dividends reduces Investment in T)
\$5,000,000 (equity income increases Investment in T)	
\$19,500,000	

- Let's compare the carrying value of A 's investment in T over A 's proportionate share of T 's equity at the end of fiscal 2024:

Typical Movements in Equity Method Investments

Equity Method Investment (Non Current Asset)	
Beginning balance (carrying value from previous period)	Sale of equity method investments (carrying value)
Purchases of equity method investments (at cost)	Dividends received from Affiliates, if any
Equity income	
Ending balance	

Coca-Cola and its Equity Method Investments

NOTE 6: EQUITY METHOD INVESTMENTS

Our consolidated net income includes our Company's proportionate share of the net income or loss of our equity method investees. When we record our proportionate share of net income, it increases equity income (loss) — net in our consolidated statement of income and our carrying value of that investment. Conversely, when we record our proportionate share of a net loss, it decreases equity income (loss) — net in our consolidated statement of income and our carrying value of that investment. The Company's proportionate share of the net income or loss of our equity method investees includes significant operating and nonoperating items recorded by our equity method investees. These items can have a significant impact on the amount of equity income (loss) — net in our consolidated statement of income and our carrying value of those investments. Refer to Note 18 for additional information related to significant operating and nonoperating items recorded by our equity method investees. The carrying values of our equity method investments are also impacted by our proportionate share of items impacting the equity method investees' AOCI.

We eliminate from our financial results all significant intercompany transactions to the extent of our ownership interest, including the intercompany portion of transactions with equity method investees.

The Company's equity method investments include, but are not limited to, our ownership interests in CCEP; Monster; AC Bebidas, S. de R.L. de C.V.; Coca-Cola FEMSA, S.A.B. de C.V.; Coca-Cola HBC AG; and Coca-Cola Bottlers Japan Holdings Inc. As of December 31, 2023, we owned 19%, 20%, 20%, 28%, 21% and 18%, respectively, of these companies' outstanding shares. As of December 31, 2023, our investments in our equity method investees in the aggregate exceeded our proportionate share of the net assets of these equity method investees by \$8,634 million. This difference is not amortized.

The Coca-Cola Company

Year Ended December 31,	2023
Net Operating Revenues	\$ 45,754 \$
Cost of goods sold	18,520
Gross Profit	27,234
Selling, general and administrative expenses	13,972
Other operating charges	1,951
Operating Income	11,311
Interest income	907
Interest expense	1,527
Equity income (loss) — net	1,691
Other income (loss) — net	570
Income Before Income Taxes	12,952
Income taxes	2,249
Consolidated Net Income	10,703
Less: Net income (loss) attributable to noncontrolling interests	(11)
Net Income Attributable to Shareowners of The Coca-Cola Company	\$ 10,714 \$
Basic Net Income Per Share¹	\$ 2.48 \$
Diluted Net Income Per Share¹	\$ 2.47 \$
Average Shares Outstanding — Basic	4,323
Effect of dilutive securities	16
Average Shares Outstanding — Diluted	4,339

Coke's Fiscal 2023 Gross Profit/Net Operating Revenues = 27,234/45,754 = 59.5%

THE COCA-COLA COMPANY AND SUBSIDIARIES
CONSOLIDATED BALANCE SHEETS
(In millions except par value)

December 31,	2023	2022
<u>ASSETS</u>		
Current Assets		
Cash and cash equivalents	\$ 9,366	\$ 9,519
Short-term investments	2,997	1,043
Total Cash, Cash Equivalents and Short-Term Investments	12,363	10,562
Marketable securities	1,300	1,069
Trade accounts receivable, less allowances of \$502 and \$516, respectively	3,410	3,487
Inventories	4,424	4,233
Prepaid expenses and other current assets	5,235	3,240
Total Current Assets	26,732	22,591
Equity method investments	19,671	18,264
Other investments	118	501
Other noncurrent assets	7,162	6,189
Deferred income tax assets	1,561	1,746
Property, plant and equipment — net	9,236	9,841
Trademarks with indefinite lives	14,349	14,214
Goodwill	18,358	18,782
Other intangible assets	516	635
Total Assets	\$ 97,703	\$ 92,763

$$\begin{aligned}
 \text{Coke's Fiscal 2023 ROA} &= [CNI + I * (1 - ETR)] / \text{Average Assets} \\
 &= [10,703 + 1527 * (1 - 0.174)] / 95,233 = 12.6\%
 \end{aligned}$$

Coke's Equity Method Investees

A summary of financial information for our equity method investees in the aggregate is as follows (in millions):

Year Ended December 31, ¹		2023	2022	2021
Net operating revenues	\$	93,862 \$	85,116 \$	79,934
Cost of goods sold		55,780	52,051	47,847
Gross profit	\$	38,082 \$	33,065 \$	32,087
Operating income	\$	11,868 \$	9,719 \$	9,089
Consolidated net income	\$	7,657 \$	6,373 \$	6,050
Less: Net income attributable to noncontrolling interests		75	102	91
Net income attributable to common shareowners	\$	7,582 \$	6,271 \$	5,959
Company equity income (loss) — net	\$	1,691 \$	1,472 \$	1,438

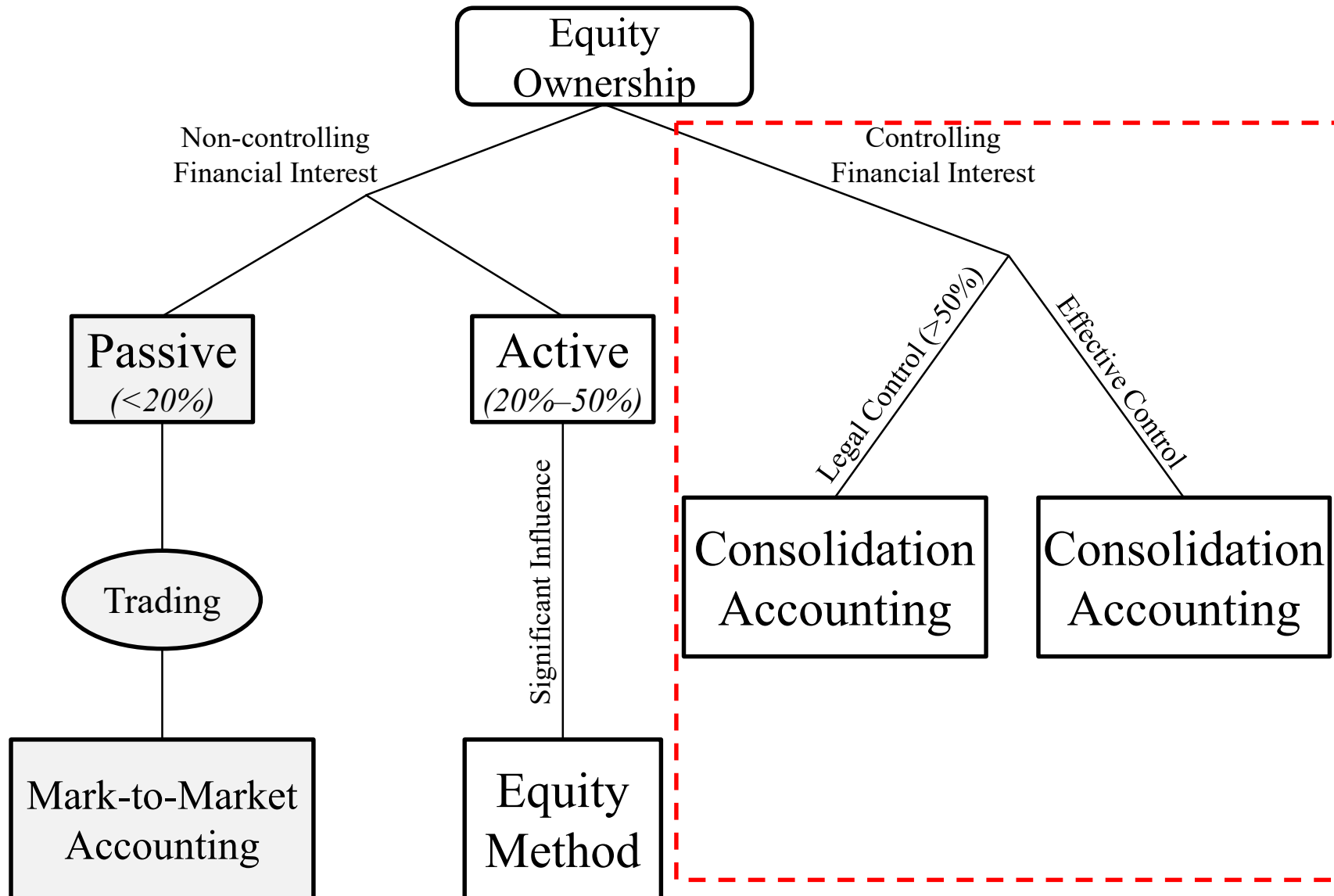
¹ The financial information represents the results of the equity method investees during the Company's period of ownership.

December 31,		2023	2022
Current assets	\$	35,087 \$	32,722
Noncurrent assets		74,464	70,523
Total assets	\$	109,551 \$	103,245
Current liabilities	\$	26,929 \$	23,580
Noncurrent liabilities		33,989	34,740
Total liabilities	\$	60,918 \$	58,320
Equity attributable to shareowners of investees	\$	47,473 \$	43,917
Equity attributable to noncontrolling interests		1,160	1,008
Total equity	\$	48,633 \$	44,925
Company equity method investments	\$	19,671 \$	18,264

$$\text{Investees' Fiscal 2023 Gross Profit/Net Operating Revenues} = 38,082/93,862 = 40.5\%$$

$$\text{Investees' Fiscal 2023 ROA} = [CNI + I * (1 - ETR)] / \text{Average Assets} = [7657 + 0] / 106,398 = 7.2\%$$

Accounting for Controlling Interests



Two Methods for Determining Control

Voting Rights Model of Control (Legal Control)

- In a voting interest entity, the equity investment is deemed sufficient to absorb the expected losses of the entity, and the equity investment has all of the characteristics of a controlling financial interest. As a result, **voting rights are the key driver** for determining which party, if any, should consolidate the entity.

Risk-and-Rewards Model of Control (Effective Control)

- A Variable Interest Entity (VIE or sometimes called an SPE) is designed in a manner where voting rights held by equity holders are ineffective in determining who has a controlling financial interest in the entity.
- Seeks to identify the **primary beneficiary**, which is the reporting enterprise that both:
 1. Has the power to direct matters that most significantly impact the economic performance of the VIE.
 2. Has the obligation to absorb losses or the right to receive the benefits of the VIE that could potentially be significant to the VIE (“losses/benefits criterion”) that absorbs the majority of the economic risks or rewards of the VIE.

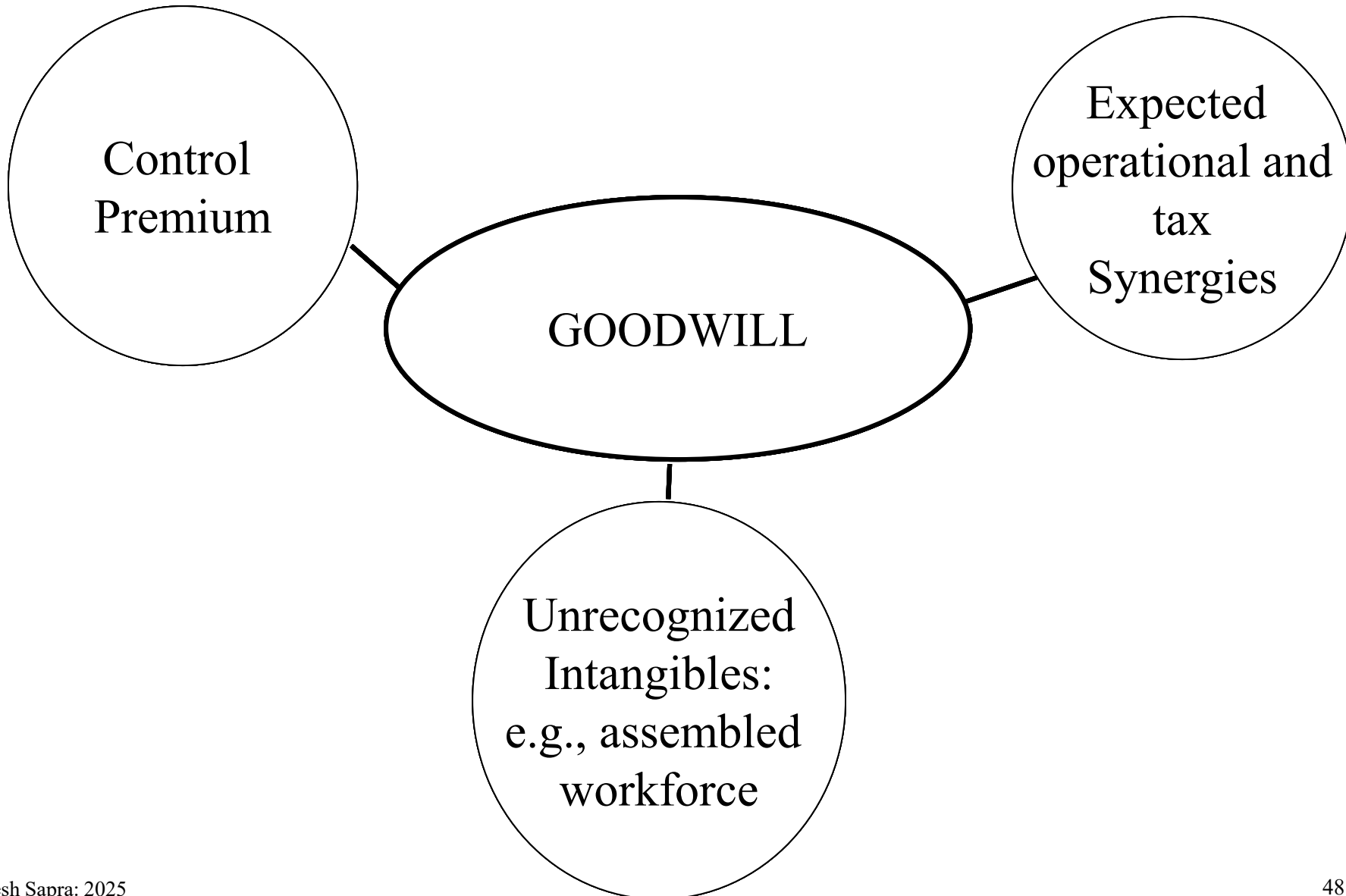
Consolidated Financial Statements

- Acquiring a controlling financial interest in a target implies that the acquirer and the target are viewed as *one reporting entity* for financial reporting purposes. The acquirer is viewed as the parent and the target as the subsidiary.
 - The parent company and subsidiary should therefore prepare one set of financial statements called *consolidated financial statements for external users*.
 - Current reporting standards require companies to use the *acquisition method* to prepare consolidated financial statements.
 - The acquisition method essentially requires an acquirer to perform a *purchase price allocation* (PPA) to explain the price paid for a business in terms on the fair value of net assets acquired.

PPA under the Acquisition Method

- Step 1: When the acquirer obtains control over the target, *regardless of the percentage of control*, the acquirer must value *100%* of the fair value of target's equity and allocate it to both controlling interests (CI) and *any* non-controlling interests (NCI).
- Step 2: The acquirer then performs a *PPA* by allocating the fair value of *100%* of the target's equity as follows:
 1. All net tangible assets of the target (tangible assets net of liabilities assumed) must be recognized and measured at *fair value*.
 2. Purchased intangible assets must be recognized and measured at fair value.
 - Purchased intangibles include IPR&D, brand names, customer lists and many others that are typically *not recognized* on the books of the target.
 3. Any *positive residual* of the excess of fair value of *100%* of target's equity over the fair value of the net identifiable assets results in Goodwill. Conversely, a *negative residual* implies a bargain purchase and results in a non-operating gain.

What is Goodwill?



Microsoft's Acquisition of Nuance Communications: Purchase Price > Fair value of Net Assets Acquired

Nuance Communications, Inc.

On March 4, 2022, we completed our acquisition of Nuance Communications, Inc. ("Nuance") for a total purchase price of \$18.8 billion, consisting primarily of cash. Nuance is a cloud and artificial intelligence ("AI") software provider with healthcare and enterprise AI experience, and the acquisition will build on our industry-specific cloud offerings. The financial results of Nuance have been included in our consolidated financial statements since the date of the acquisition. Nuance is reported as part of our Intelligent Cloud segment.

The allocation of the purchase price to goodwill was completed as of December 31, 2022. The major classes of assets and liabilities to which we have allocated the purchase price were as follows:

(In millions)

Goodwill ^(a)	\$	16,326
Intangible assets		4,365
Other assets		42
Other liabilities ^(b)		(1,972)
Total	\$	<u>18,761</u>

Following are the details of the purchase price allocated to the intangible assets acquired:

(In millions, except average life)	Amount	Weighted Average Life
Customer-related	\$ 2,610	9 years
Technology-based	1,540	5 years
Marketing-related	215	4 years
Total	<u>\$ 4,365</u>	<u>7 years</u>

Intangible Assets \$4,365

Goodwill 16,326

Other Assets 42

Other Liabilities \$1,972

Cash 18,761

Suppose Microsoft had, instead, purchased only 80% of Nuance's outstanding shares for \$15,000 millions in cash...

- Fair value of 100% of Nuance's equity = Fair value of Microsoft's CI of \$15,000 million + Fair value of NCI.
- How do we value the NCI (i.e., equity stake that was not acquired)?
 - Revalue 100% of Nuance's equity by using Microsoft's valuation of its CI = $\$15,000 \text{ million} / 0.8 = \$18,750 \text{ million}$ so that the fair value of NCI = $\$18,750 \text{ million} * 0.20 = \$3,750 \text{ million}$.
- Microsoft then accounts for the acquisition of 80% Nuance's equity (all \$ amounts below in millions):

Intangible Assets	\$4,365
Goodwill	16,326
Other Assets	42

Other Liabilities	\$1,972
Cash	15,000
NCI (Equity)	3,750

JP Morgan's Acquisition of Signature Bank In May 2023: Purchase Price < Fair Value of Net Assets Acquired

Purchase price consideration	\$	67,834
Assets		
Securities	\$	30,285
Loans ^(c)		153,242
Core deposit and customer relationship intangibles		1,455
Indemnification assets - Shared-loss agreements		675
Accounts receivable and other assets ^{(c)(d)}		6,574
Total assets acquired	\$	192,231
Liabilities		
Deposits	\$	87,572
FHLB advances		27,919
Lending-related commitments		2,614
Accounts payable and other liabilities ^{(c)(d)}		2,793
Deferred tax liabilities		724
Total liabilities assumed	\$	121,622
Fair value of net assets acquired	\$	70,609
Estimated gain on acquisition, after-tax	\$	2,775

Assets 192,231

Liabilities \$121,622

Gain 2,775

Cash & Debt 67,834

Goodwill Impairment

- Goodwill is *not amortized* for financial reporting purposes. Rather, it is tested for impairment at least once annually.
- The fair value (FV_{GW}) of goodwill is calculated at the time of impairment testing and compared to its carrying value (CV_{GW}).
 - If $FV_{GW} < CV_{GW}$, Impairment Loss = $CV_{GW} - FV_{GW}$
Impairment Loss \$ $CV_{GW} - FV_{GW}$
Provision for Impairment \$ $CV_{GW} - FV_{GW}$
 - Otherwise, goodwill is not impaired.